

Engineering Mathematics

500+ One-Liners for Rapid Revision

IBPS SO (IT Officer) Examination - Complete Preparation Series

100% Syllabus Coverage - MCQ-Oriented - Zero Filler

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Table of Contents

Engineering Mathematics - 500+ One-Liners	4
1. Mathematical Logic	4
Propositional Logic	4
Tautology, Contradiction, Contingency	4
Logical Equivalences	4
Normal Forms	4
Rules of Inference	4
Predicate Logic	5
2. Set Theory and Relations	5
Sets and Power Sets	5
Inclusion-Exclusion	5
Relations	5
Functions	5
Countability	5
3. Partial Orders and Lattices	5
4. Algebraic Structures	6
5. Graph Theory	6
6. Combinatorics	6
7. Matrices and Determinants	7
8. System of Linear Equations	7
9. Eigenvalues and Eigenvectors	7
10. LU Decomposition	8
11. Limits, Continuity, Differentiability	8
12. Applications of Derivatives	8
13. Integration	8
14. Probability	9

15. Probability Distributions	9
16. Statistics	9
17. Numerical Solutions of Equations	10
18. Numerical Integration	10
19. Numerical Solution of ODEs	10
20. Exam Focus Rapid-Fire	10
21. Extended Rapid-Fire: Logic and Sets	11
22. Extended Rapid-Fire: Lattices and Algebra	11
23. Extended Rapid-Fire: Graphs	11
24. Extended Rapid-Fire: Combinatorics	12
25. Extended Rapid-Fire: Linear Algebra	12
26. Extended Rapid-Fire: Calculus	13
27. Extended Rapid-Fire: Probability	13
28. Extended Rapid-Fire: Distributions and Statistics	14
29. Extended Rapid-Fire: Numerical Methods	14
30. Extended Rapid-Fire: Cross-Topic Mix	15
31. Extended Rapid-Fire: More Logic and Proof	15
32. Extended Rapid-Fire: More Sets and Relations	15
33. Extended Rapid-Fire: More Linear Algebra and Numerical	16
34. Extended Rapid-Fire: More Calculus and Series	16
35. Extended Rapid-Fire: More Probability and Stats	16
36. Extended Rapid-Fire: Final Mixed Set	17
37. Bonus One-Liners	17

Engineering Mathematics - 500+ One-Liners

- This rapid-revision book packs **500+** one-liners across **5** pillars: **Discrete Math**, **Linear Algebra**, **Calculus**, **Probability/Statistics**, and **Numerical Methods**.

1. Mathematical Logic

Propositional Logic

- There are **5** basic connectives: **NOT ($\neg$)**, **AND ($\wedge$)**, **OR ($\vee$)**, **implication ($\rightarrow$)**, and **biconditional ($\leftrightarrow$)**.
- A truth table for **n** variables has exactly **2^n** rows.
- Implication $p \rightarrow q$** is false in only **1** case: when **p is true and q is false**.
- Conjunction $p \wedge q$** is true in only **1** of **4** rows: when **both** are true.
- Disjunction $p \vee q$** is false in only **1** row: when **both** are false.
- Biconditional $p \leftrightarrow q$** is true in exactly **2** rows: when **both truth values match**.
- The **XOR** operator is true in exactly **2** of **4** rows: when **operands differ**.
- The implication equivalence is **$p \rightarrow q \equiv \neg p \vee q$** , using **1** negation and **1** OR.
- The **1** equivalent form of an implication is its **contrapositive $\neg q \rightarrow \neg p$** .
- There are **2** non-equivalent transforms of $p \rightarrow q$: its **converse $q \rightarrow p$** and **inverse $\neg p \rightarrow \neg q$** .
- A **vacuously true** implication occurs whenever the premise is **false**, giving value **true**.

Tautology, Contradiction, Contingency

- There are **3** classifications of a formula: **tautology**, **contradiction**, and **contingency**.
- A **tautology** is true in all **2^n** rows, e.g., **$p \vee \neg p$** .
- A **contradiction** is false in all rows, e.g., **$p \wedge \neg p$** .
- A **contingency** has at least **1** true and **1** false row.
- A formula is **satisfiable** if it is true in at least **1** row.

Logical Equivalences

- De Morgan's** 2 laws are **$\neg(p \wedge q) \equiv \neg p \vee \neg q$** and **$\neg(p \vee q) \equiv \neg p \wedge \neg q$** .
- The **2** identity laws are **$p \wedge T \equiv p$** and **$p \vee F \equiv p$** .
- The **2** domination laws are **$p \vee T \equiv T$** and **$p \wedge F \equiv F$** .
- The **2** idempotent laws are **$p \vee p \equiv p$** and **$p \wedge p \equiv p$** .
- The **absorption law** states **$p \vee (p \wedge q) \equiv p$** .

Normal Forms

- There are **2** canonical normal forms: **CNF** (AND of ORs, product of sums) and **DNF** (OR of ANDs, sum of products).
- DNF** is a **sum of products (SOP)**; **CNF** is a **product of sums (POS)**.

Rules of Inference

- Modus Ponens** uses **2** premises **p** and **$p \rightarrow q$** to infer **q**.
- Modus Tollens** uses **$\neg q$** and **$p \rightarrow q$** to infer **$\neg p$** .
- Hypothetical syllogism** chains **$p \rightarrow q$** and **$q \rightarrow r$** into **$p \rightarrow r$** .
- Disjunctive syllogism** uses **$p \vee q$** and **$\neg p$** to infer **q**.
- There are **2** classic fallacies: **affirming the consequent** and **denying the antecedent**.

Predicate Logic

- There are **2** quantifiers: **universal** $\forall$ ("for all") and **existential** $\exists$ ("there exists").
- Negation flips quantifiers: $\neg \forall x P \equiv \exists x \neg P$ and $\neg \exists x P \equiv \forall x \neg P$.
- Quantifier order matters: $\forall x \exists y$ is weaker than $\exists y \forall x$.

2. Set Theory and Relations

Sets and Power Sets

- A set of **n** elements has **2^n** subsets and **$2^n - 1$** proper subsets.
- The **power set** of an n-element set has **2^n** members.
- The **empty set** $\emptyset$ is a subset of **every** set and has exactly **1** subset (itself).
- The set $\{\emptyset\}$ has exactly **1** element, distinct from $\emptyset$.
- There are **4** core set operations: **union**, **intersection**, **difference**, and **complement**.
- The number of size-r subsets of an n-set is **$C(n,r)$** .

Inclusion-Exclusion

- The 2-set formula is **$|A \cup B| = |A| + |B| - |A \cap B|$** .
- The 3-set formula adds **3** singles, subtracts **3** pairs, and adds **1** triple.
- The Cartesian product satisfies **$|A \times B| = |A| \cdot |B|$** .

Relations

- A relation on n elements is any of **$2^{(n^2)}$** subsets of **$A \times A$** .
- There are **4** key relation properties: **reflexive**, **symmetric**, **antisymmetric**, **transitive**.
- The count of **reflexive** relations on n elements is **$2^{(n^2-n)}$** .
- The count of **symmetric** relations on n elements is **$2^{(n(n+1)/2)}$** .
- An **equivalence relation** satisfies **3** properties: **reflexive**, **symmetric**, **transitive**.
- An equivalence relation **partitions** a set into **disjoint** equivalence classes.
- The number of equivalence relations on n elements equals the **Bell number B(n)**.
- A **partial order** satisfies **3** properties: **reflexive**, **antisymmetric**, **transitive**.

Functions

- There are **3** function types: **injective** (one-one), **surjective** (onto), **bijective** (both).
- Total functions from an m-set to an n-set number **n^m** .
- Injective functions from m to n ($m \leq n$) number **$n!/(n-m)!$** .
- A **bijection** is the only type with an **inverse**.
- Composition **$(f \circ g)(x) = f(g(x))$** applies **g first**.

Countability

- **$\mathbb{Z}$ and $\mathbb{Q}$** are **countable**; **$\mathbb{R}$** is **uncountable**.

3. Partial Orders and Lattices

- A **POSET** requires **3** properties: **reflexive**, **antisymmetric**, **transitive**.
- A **lattice** requires that every pair have both a **LUB (join)** and **GLB (meet)**.

- A **bounded lattice** has **2** special elements: a least **0** and greatest **1**.
- A **Boolean algebra** is a lattice with **3** traits: **bounded**, **distributive**, **complemented**.
- Every **chain** (total order) is automatically a **distributive** lattice.
- The **LUB** is the **least upper bound**; the **GLB** is the **greatest lower bound**.

4. Algebraic Structures

- A **semigroup** needs **2** properties: **closure** and **associativity**.
- A **monoid** adds **1** property to a semigroup: an **identity element**.
- A **group** needs **4** properties: **closure**, **associativity**, **identity**, **inverse**.
- An **abelian group** adds **1** property: **commutativity**.
- **Lagrange's theorem** says a subgroup's order **divides** the group's order.
- Every **cyclic group** is **abelian** and generated by **1** element.
- The **order of an element a** is the smallest k with $a^k = e$.

5. Graph Theory

- The **handshaking lemma** states $\sum \deg(v) = 2|E|$.
- The number of **odd-degree** vertices in any graph is always **even**.
- A **complete graph** K_n has $n(n-1)/2$ edges.
- A **complete bipartite** $K_{m,n}$ has $m \cdot n$ edges.
- A **tree** on n vertices has exactly $n-1$ edges.
- A graph is **bipartite** iff it has **no odd-length cycle**.
- An **Eulerian circuit** exists iff the graph is connected and all vertices have **even** degree.
- An **Eulerian path** exists iff exactly **0 or 2** vertices have **odd** degree.
- A **Hamiltonian** cycle visits every **vertex** once and has **no** simple necessary-sufficient condition.
- **Euler's formula** for connected planar graphs is $V - E + F = 2$.
- A simple planar graph satisfies $E \leq 3V - 6$; bipartite planar satisfies $E \leq 2V - 4$.
- The **2** smallest non-planar graphs are K_5 and $K_{3,3}$ (Kuratowski).
- The **chromatic number** of K_n is n , of a bipartite graph is **2**.
- An **odd cycle** has chromatic number **3**; an **even cycle** has **2**.
- The **Four Color Theorem** says every planar graph needs at most **4** colors.
- The number of spanning trees of K_n is n^{n-2} (Cayley).
- Adding one edge to a tree creates exactly **1** cycle.

6. Combinatorics

- The **2** basic counting rules are the **sum rule** (+) and **product rule** ($\times$).
- A **permutation** $P(n,r) = n!/(n-r)!$ counts arrangements where **order matters**.
- A **combination** $C(n,r) = n!/(r!(n-r)!)$ counts selections where **order does not matter**.
- **Circular permutations** of n objects number $(n-1)!$.
- The **pigeonhole principle** guarantees some box holds at least $\lceil n/k \rceil$ items.
- The **binomial theorem** expands $(x+y)^n = \sum C(n,k) x^{n-k} y^k$.
- The sum of binomial coefficients for power n is 2^n .
- A linear homogeneous recurrence is solved via its **characteristic equation**.
- Distinct roots give $a_n = A r_1^n + B r_2^n$; a repeated root gives $(A+Bn) r^n$.

- The number of **derangements** is $D(n) = n! \sum (-1)^k/k!$.
- The basic generating function is $1/(1-x) = \sum x^n$.

7. Matrices and Determinants

- The **identity matrix** I satisfies $AI = IA = A$ with **1s** on the diagonal.
- A **symmetric** matrix satisfies $A = A^T$; a **skew-symmetric** satisfies $A = -A^T$.
- A skew-symmetric matrix has all **0s** on its diagonal.
- An odd-order skew-symmetric matrix has determinant **0**.
- An **orthogonal** matrix satisfies $AA^T = I$, so $A^{-1} = A^T$ and $\det = \pm 1$.
- Matrix multiplication is **associative** and **distributive** but **not commutative**.
- The product transpose rule is $(AB)^T = B^T A^T$ (reversed order).
- The product inverse rule is $(AB)^{-1} = B^{-1} A^{-1}$ (reversed order).
- Naive $n \times n$ matrix multiplication costs $O(n^3)$; Strassen costs $O(n^{2.81})$.
- A 2×2 determinant equals $ad - bc$.
- $\det(A^T) = \det(A)$ and $\det(AB) = \det(A) \cdot \det(B)$.
- Scaling an $n \times n$ matrix gives $\det(kA) = k^n \det(A)$.
- Swapping **2** rows multiplies the determinant by **-1**.
- A matrix with **2** identical rows has determinant **0**.
- The inverse formula is $A^{-1} = \text{adj}(A)/\det(A)$, valid iff $\det \neq 0$.
- The product $A \cdot \text{adj}(A) = \det(A) \cdot I$.
- For an $n \times n$ matrix, $\det(\text{adj } A) = \det(A)^{n-1}$.
- The **rank** of a matrix is the number of **linearly independent rows** (= columns).
- Rank equals the number of **non-zero rows** in row-echelon form.
- The only matrix with rank **0** is the **zero matrix**.
- There are **3** elementary row operations: **row swap**, **scaling**, **add-multiple-of-row**.
- Elementary row operations preserve **rank** and the **solution set**.

8. System of Linear Equations

- A **homogeneous** system $Ax=0$ always has the **trivial** solution.
- For $Ax=b$, the system is **inconsistent** when $\text{rank}(A) < \text{rank}([A|b])$.
- A **unique** solution exists when $\text{rank}(A) = \text{rank}([A|b]) = n$.
- **Infinite** solutions occur when $\text{rank}(A) = \text{rank}([A|b]) < n$, with $n - \text{rank}$ free variables.
- Homogeneous $Ax=0$ has non-trivial solutions iff $\det(A) = 0$.
- **Gaussian elimination** reduces to **upper triangular** form then back-substitutes, costing $O(n^3)$.
- **Cramer's rule** gives $x_i = \det(A_i)/\det(A)$ and fails when $\det(A)=0$.

9. Eigenvalues and Eigenvectors

- The eigenvalue equation is $Av = \lambda v$ with $v \neq 0$.
- Eigenvalues solve the **characteristic equation** $\det(A - \lambda I) = 0$.
- The **sum** of eigenvalues equals the **trace** of A .
- The **product** of eigenvalues equals the **determinant** of A .
- A **triangular** matrix's eigenvalues are its **diagonal entries**.
- **Symmetric** matrices have all **real** eigenvalues.

- Eigenvalues of A^{-1} are $1/\lambda$; of A^k are λ^k .
- A **singular** matrix has **0** as an eigenvalue.
- The **Cayley-Hamilton theorem** says every matrix satisfies its own **characteristic equation**.
- Diagonalization writes $A = PDP^{-1}$, valid iff A has **n** independent eigenvectors.

10. LU Decomposition

- **LU decomposition** factors $A = LU$ into **lower** and **upper** triangular matrices.
- **Doolittle** puts unit diagonal on **L**; **Crout** puts it on **U**.
- LU solves $Ly=b$ by forward substitution then $Ux=y$ by back substitution.
- LU factorization costs $O(n^3)$; each subsequent solve costs $O(n^2)$.
- LU is preferred for **multiple right-hand sides** since L, U are reused.

11. Limits, Continuity, Differentiability

- The standard limit $\lim (\sin x)/x = 1$ as $x \rightarrow 0$.
- The standard limit $\lim (1 - \cos x)/x^2 = 1/2$ as $x \rightarrow 0$.
- The standard limit $\lim (e^x - 1)/x = 1$ as $x \rightarrow 0$.
- The standard limit $\lim (1 + 1/n)^n = e$ as $n \rightarrow \infty$.
- **L'Hopital's rule** applies only to $0/0$ or ∞/∞ forms: $\lim f/g = \lim f'/g'$.
- f is **continuous** at a iff $\lim f(x) = f(a)$.
- **Differentiability** implies **continuity**, but not the reverse.
- The function $|x|$ is continuous but **not differentiable** at $x = 0$.
- The **product rule** is $(uv)' = u'v + uv'$.
- The **quotient rule** is $(u/v)' = (u'v - uv')/v^2$.
- The **chain rule** is $d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$.
- The power derivative is $d/dx x^n = n x^{n-1}$.

12. Applications of Derivatives

- **Critical points** occur where $f'(x) = 0$ or f' is undefined.
- The **second derivative test**: $f'' > 0$ is a minimum, $f'' < 0$ a maximum, $f'' = 0$ inconclusive.
- An **inflection point** occurs where $f'' = 0$ and concavity changes.
- **Global extrema** on $[a, b]$ require checking critical points **and endpoints**.
- **Rolle's theorem** guarantees $f'(c) = 0$ when $f(a) = f(b)$.
- **Lagrange's MVT** guarantees $f'(c) = (f(b) - f(a))/(b - a)$.
- **Rolle's** is the special case of **Lagrange's** when $f(a) = f(b)$.

13. Integration

- The power integral is $\int x^n dx = x^{n+1}/(n+1) + C$ for $n \neq -1$.
- The reciprocal integral is $\int (1/x) dx = \ln|x| + C$.
- **Integration by parts** is $\int u dv = uv - \int v du$.
- The u -choice order in by-parts follows **ILATE**: Inverse, Log, Algebraic, Trig, Exponential.
- The **Fundamental Theorem** gives $\int_a^b f = F(b) - F(a)$.
- The **king property** is $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

- Area under a curve uses $\int_a^b |f(x)| dx$ to avoid sign cancellation.

14. Probability

- The probability axioms require $0 \leq P(A) \leq 1$ and $P(S) = 1$.
- The general addition rule is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional probability** is $P(A|B) = P(A \cap B)/P(B)$.
- The **multiplication rule** is $P(A \cap B) = P(A|B) \cdot P(B)$.
- Independent** events satisfy $P(A \cap B) = P(A) \cdot P(B)$.
- Mutually exclusive** non-zero events are always **dependent**.
- Total probability** is $P(A) = \sum P(A|B_i)P(B_i)$.
- Bayes' theorem** computes $P(B_i|A) = P(A|B_i)P(B_i)/\sum P(A|B_j)P(B_j)$.
- There are **2** random variable types: **discrete** (PMF) and **continuous** (PDF).
- A **PMF** sums to **1**; a **PDF** integrates to **1**.
- For a **continuous** RV, $P(X = a) = 0$.
- The **CDF** is $F(x) = P(X \leq x)$, non-decreasing from **0** to **1**.
- Expectation** is $E[X] = \sum x P(x)$ or $\int x f(x) dx$.
- Variance** is $\text{Var}(X) = E[X^2] - (E[X])^2$.
- Linear transform variance is $\text{Var}(aX+b) = a^2 \text{Var}(X)$.
- Expectation is linear: $E[X+Y] = E[X]+E[Y]$ always.
- $\text{Var}(X+Y) = \text{Var}(X)+\text{Var}(Y)$ only when X,Y are **independent**.

15. Probability Distributions

- The **binomial** PMF is $C(n,k) p^k (1-p)^{n-k}$ with mean **np** and variance **np(1-p)**.
- The **Poisson** PMF is $e^{-\lambda} \lambda^k / k!$ with mean = variance = λ .
- Poisson approximates binomial when **n large, p small, np = λ** .
- The **continuous uniform** on [a,b] has mean $(a+b)/2$ and variance $(b-a)^2/12$.
- The **normal** distribution has parameters mean μ and variance σ^2 .
- The **standard normal** $Z = (X-\mu)/\sigma$ has mean **0** and variance **1**.
- The **empirical rule** gives **68%, 95%, 99.7%** within 1, 2, 3 standard deviations.
- The **exponential** PDF is $\lambda e^{-\lambda x}$ with mean $1/\lambda$ and variance $1/\lambda^2$.
- The exponential distribution is **memoryless**: $P(X>s+t|X>s)=P(X>t)$.
- A normal distribution is **symmetric** with **mean = median = mode**.

16. Statistics

- The **3** central tendency measures are **mean, median, and mode**.
- The **mean** is sensitive to outliers; the **median** is robust.
- The empirical mode relation is **Mode = 3·Median – 2·Mean**.
- Population variance** divides by **N**; **sample variance** divides by **n-1**.
- The **range** is **max – min**; **standard deviation** is $\sqrt{\text{variance}}$.
- The **coefficient of variation** is σ/μ .
- Pearson's r** lies in **[-1, 1]** and equals $\text{cov}(X,Y)/(\sigma_X \sigma_Y)$.
- r = 0** indicates no **linear** correlation.
- The regression slope is **b = cov(X,Y)/Var(X)**.

- The correlation equals the **geometric mean** of the **2** regression coefficients: $r = \sqrt{(b_{xy} \cdot b_{yx})}$.

17. Numerical Solutions of Equations

- The **bisection** method needs $f(a) \cdot f(b) < 0$ and has **linear** convergence (order **1**).
- Bisection error after n steps is $\leq (b-a)/2^n$.
- **Newton-Raphson** iterates $x - f(x)/f'(x)$ with **quadratic** convergence (order **2**).
- Newton-Raphson fails when $f'(x) = 0$.
- The **secant** method has convergence order ≈ 1.618 .
- **Regula-falsi** keeps a **bracketing** interval, guaranteeing convergence.
- **Bisection** and **regula-falsi** are the **2** bracketing methods that always converge.
- **Gauss elimination** is a **direct** method costing $O(n^3)$.
- **Gauss-Seidel** converges when the matrix is **diagonally dominant**.
- **Gauss-Seidel** uses **updated** values immediately, unlike **Jacobi**.
- **Absolute error** is $|true - approx|$; **relative error** divides by $|true|$.

18. Numerical Integration

- The **trapezoidal rule** has error $O(h^2)$ and is exact for degree **1** polynomials.
- **Simpson's 1/3 rule** has error $O(h^4)$ and is exact for degree **3** polynomials.
- **Simpson's 1/3** requires an **even** number of subintervals.
- The trapezoidal formula is $(h/2)[f_0 + 2\sum f_{mid} + f_n]$.
- Simpson's 1/3 weights are **1, 4, 2, 4, ..., 1** scaled by $h/3$.
- **Simpson's 3/8 rule** also has error $O(h^4)$ and is exact to degree **3**.

19. Numerical Solution of ODEs

- **Euler's method** is $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, first-order with error $O(h)$.
- **RK4** is $y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$, order **4**.
- RK4 evaluates the slope **4** times per step, weighting middle slopes by **2**.
- **Multi-step** methods like **Adams-Bashforth** use **several** previous points.
- A **predictor-corrector** pairs an **explicit** predictor with an **implicit** corrector.

20. Exam Focus Rapid-Fire

- Test logical equivalence using **truth tables** or the **De Morgan / distributive** laws.
- Classify relations: **RST** gives an **equivalence**, **RAT** gives a **partial order**.
- Confirm a lattice by checking **LUB** and **GLB** exist for **all** pairs.
- Decide counting type: **order matters** is permutation, otherwise **combination**.
- Solve recurrences via the **characteristic equation**, using $(A+Bn)r^n$ for repeated roots.
- Use **row-echelon** form to find rank and $A^{-1} = \text{adj}(A)/\det(A)$ for inverse.
- Apply **trace** = $\sum \lambda$ and **det** = $\prod \lambda$ as quick eigenvalue checks.
- Use **Bayes** when conditioning must be **reversed**.
- Remember **Newton-Raphson** order **2**, **trapezoidal** $O(h^2)$, **Simpson** $O(h^4)$.
- Apply **handshaking** $\sum \deg = 2E$ and **Euler** $V - E + F = 2$ for graph counting questions.

21. Extended Rapid-Fire: Logic and Sets

- The **commutative** laws hold for $\wedge$ and $\vee$ but **not** for $\rightarrow$.
- The **associative** law lets $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ regroup freely.
- The **distributive** law gives $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- The **2** negation laws are $\neg\neg p \equiv p$ (double negation) and $\neg T \equiv F$.
- A formula with **k** clauses in CNF is **satisfiable** if all clauses can be **simultaneously true**.
- The **addition** rule infers $p \vee q$ from a single p .
- The **simplification** rule infers p from $p \wedge q$.
- The **conjunction** rule infers $p \wedge q$ from **2** premises p and q .
- The **resolution** rule combines $p \vee q$ and $\neg p \vee r$ into $q \vee r$.
- A **predicate** becomes a proposition after **quantification** or **value assignment**.
- The statement "**some**" maps to $\exists$; "**every/all**" maps to $\forall$.
- Negating "all swans are white" gives "**there exists a swan that is not white**".
- The set **difference** satisfies $A - B = A \cap B'$.
- **Symmetric difference** $A \Delta B = (A - B) \cup (B - A)$ contains elements in exactly **1** set.
- Two sets are **disjoint** when $A \cap B = \emptyset$.
- The **universal set U** satisfies $A \cup A' = U$ and $A \cap A' = \emptyset$.
- **De Morgan** for sets: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
- The number of **onto** functions from a 3-set to a 2-set is **6**.
- A relation that is both **symmetric and antisymmetric** contains only **diagonal** pairs.
- An **irreflexive** relation contains **no** pair of the form (a, a) .
- A **total (linear) order** adds **comparability** to a partial order.
- The **transitive closure** adds the minimum pairs to make a relation **transitive**.
- A function from an n-set to itself that is injective is automatically **surjective** (and bijective).
- The number of **bijections** on an n-set is **n!**.

22. Extended Rapid-Fire: Lattices and Algebra

- The **join** of two elements is their **LUB**; the **meet** is their **GLB**.
- In a **distributive** lattice, complements (if they exist) are **unique**.
- A **sublattice** is closed under both **meet** and **join**.
- The divisibility relation on positive integers forms a **lattice** with meet **gcd** and join **lcm**.
- The power set under $\subseteq$ forms a **Boolean lattice** with 2^n elements.
- The **identity** element of addition is **0**; of multiplication is **1**.
- A **ring** has **2** operations with addition forming an **abelian group**.
- A **field** requires every **non-zero** element to have a multiplicative **inverse**.
- The set of integers under addition is an **abelian group** but under multiplication only a **monoid**.
- The **order** of the group $(\mathbb{Z}_n, +)$ is **n**.
- A group with **prime** order is always **cyclic**.
- The **identity** of a group is **unique**, and each element has a **unique** inverse.

23. Extended Rapid-Fire: Graphs

- A **walk** allows repeats; a **trail** repeats no edge; a **path** repeats no vertex.
- A graph with **n** vertices can have at most **$n(n-1)/2$** edges if simple.

- A **regular** graph of degree r on n vertices has $nr/2$ edges.
- The **minimum** number of edges in a connected graph on n vertices is $n-1$.
- A **forest** with k components and n vertices has $n-k$ edges.
- A **cut vertex** disconnects the graph when removed.
- A **bridge** is an edge whose removal increases the number of **components**.
- The **complement** of a graph on n vertices and e edges has $n(n-1)/2 - e$ edges.
- A **self-complementary** graph on n vertices needs $n \equiv 0 \text{ or } 1 \pmod{4}$.
- A **tournament** is a complete graph with **directed** edges.
- The **adjacency matrix** of an undirected graph is **symmetric**.
- The (i,j) entry of A^k counts walks of length k from i to j .
- A **tree** with n vertices has exactly $n-1$ edges and 0 cycles.
- A **binary tree** of height h has at most $2^{h+1} - 1$ nodes.
- A graph is a **tree** iff it is **connected** and has $n-1$ edges.
- The **degree** of every vertex in a cycle C_n is 2 .
- The **Petersen graph** is a famous **non-planar**, **3-regular** graph.
- A planar graph divides the plane into F regions counting the **outer** face.

24. Extended Rapid-Fire: Combinatorics

- Arranging n objects with repeats $n_1, n_2, \dots$ gives $n!/(n_1! n_2! \dots)$.
- Selecting r from n **with repetition** (combinations) gives $C(n+r-1, r)$.
- The number of binary strings of length n is 2^n .
- The number of n -bit strings with exactly k ones is $C(n, k)$.
- The **Catalan number** $C_n = C(2n, n)/(n+1)$ counts balanced parentheses.
- The number of ways to seat n people around a circular table is $(n-1)!$.
- **Pascal's identity** is $C(n, k) = C(n-1, k-1) + C(n-1, k)$.
- The middle binomial coefficient $C(n, n/2)$ is the **largest**.
- The **Fibonacci** recurrence is $F_n = F_{n-1} + F_{n-2}$ with characteristic root **golden ratio** ϕ .
- The number of subsets of even size of an n -set is 2^{n-1} .
- The **pigeonhole** principle guarantees a repeat among $n+1$ items in n categories.
- A recurrence with characteristic roots **2 and 3** has solution $A \cdot 2^n + B \cdot 3^n$.

25. Extended Rapid-Fire: Linear Algebra

- A matrix is **singular** iff its determinant is 0 .
- The **trace** is the sum of **diagonal** entries and is **invariant** under similarity.
- The **transpose** satisfies $(A^T)^T = A$ and $(A+B)^T = A^T + B^T$.
- A **nilpotent** matrix has all eigenvalues 0 .
- An **idempotent** matrix satisfies $A^2 = A$ with eigenvalues 0 or 1 .
- An **involutory** matrix satisfies $A^2 = I$ with eigenvalues ± 1 .
- The rank of an $m \times n$ matrix is at most $\min(m, n)$.
- The **nullity** plus the **rank** equals the number of **columns** (rank-nullity).
- A **diagonal** matrix's inverse inverts each **diagonal** entry.
- The determinant of a **diagonal/triangular** matrix is the **product** of diagonal entries.
- The number of **pivots** in row-echelon form equals the **rank**.
- **Similar** matrices share the same **eigenvalues**, **trace**, and **determinant**.

- A **positive definite** matrix has all **positive** eigenvalues.
- The **characteristic polynomial** of an $n \times n$ matrix has degree **n** .
- For a 2×2 matrix, eigenvalues solve **$\lambda^2 - (\text{trace})\lambda + \det = 0$** .
- A **real symmetric** matrix is always **diagonalizable** by an orthogonal matrix.
- The **condition number** measures sensitivity of **$Ax=b$** to perturbations.
- An **augmented matrix** appends the constant vector **b** as an extra column.

26. Extended Rapid-Fire: Calculus

- The **derivative** of **$\sin x$** is **$\cos x$** , of **$\cos x$** is **$-\sin x$** .
- The **derivative** of **$\tan x$** is **$\sec^2 x$** .
- The **derivative** of **$\ln x$** is **$1/x$** , of **e^x** is **e^x** .
- The **derivative** of **a^x** is **$a^x \ln a$** .
- The **n th derivative** of **e^x** is still **e^x** .
- A function is **increasing** where **$f' > 0$** and decreasing where **$f' < 0$** .
- A function is **concave up** where **$f'' > 0$** .
- The **tangent** line at $x=a$ has slope **$f'(a)$** .
- The limit **$\lim (\tan x)/x = 1$** as $x \rightarrow 0$.
- The limit **$\lim (\ln(1+x))/x = 1$** as $x \rightarrow 0$.
- A **removable** discontinuity occurs when the limit **exists** but $\neq f(a)$.
- A **jump** discontinuity has unequal **left and right** limits.
- The **Squeeze theorem** bounds a function between **2** others with equal limits.
- **$\int \sec^2 x \, dx = \tan x + C$** and **$\int (1/(1+x^2)) \, dx = \arctan x + C$** .
- **$\int (1/\sqrt{1-x^2}) \, dx = \arcsin x + C$** .
- The integral **$\int_0^\infty e^{-x} \, dx = 1$** .
- A **definite integral** of an **odd** function over **$[-a, a]$** is **0**.
- A **definite integral** of an **even** function over **$[-a, a]$** is **$2 \int_0^a$** .
- The **average value** of f on $[a, b]$ is **$(1/(b-a)) \int_a^b f \, dx$** .
- The **area between two curves** is **$\int (\text{top} - \text{bottom}) \, dx$** .
- An **improper integral** has an infinite **limit** or unbounded **integrand**.
- The **Gamma function** satisfies **$\Gamma(n) = (n-1)!$** for positive integers.

27. Extended Rapid-Fire: Probability

- The probability of the **complement** is **$P(A') = 1 - P(A)$** .
- The probability of the **impossible** event is **0**; of the **certain** event is **1**.
- For **equally likely** outcomes, **$P(A) = \text{favorable}/\text{total}$** .
- Rolling 2 dice has **36** outcomes; sum **7** has probability **$6/36 = 1/6$** .
- A standard deck has **52** cards, **4** suits, and **13** ranks.
- Drawing an **ace** from a deck has probability **$4/52 = 1/13$** .
- The **odds** in favor are **$P(A) : P(A')$** .
- **Mutually exclusive** events satisfy **$P(A \cap B) = 0$** .
- **Pairwise independence** does **not** imply **mutual independence**.
- The **expected value** of a fair die is **3.5**.
- The **variance** of a single fair die is **35/12**.
- A **Bernoulli** trial has **2** outcomes with success probability **p** .

- The sum of n independent Bernoulli(p) trials is **Binomial(n, p)**.
- The **covariance** of independent variables is **0**.
- **Chebyshev's inequality** bounds $P(|X - \mu| \geq k\sigma) \leq 1/k^2$.
- The **law of large numbers** says the sample mean converges to μ .
- The **central limit theorem** says sums of many independent variables approach a **normal** distribution.

28. Extended Rapid-Fire: Distributions and Statistics

- The **geometric** distribution models trials until the **first success**, mean $1/p$.
- The **geometric** distribution is the only **discrete** memoryless distribution.
- The **hypergeometric** distribution models sampling **without replacement**.
- The **binomial** models sampling **with replacement** (independent trials).
- The **Poisson** models **rare** events over a fixed interval.
- The **standard normal** table gives $P(Z \leq z)$ for the cumulative area.
- About **50%** of a normal distribution lies **below the mean**.
- The **mode** of a normal distribution equals its **mean**.
- A **right-skewed** distribution has **mean > median > mode**.
- A **left-skewed** distribution has **mean < median < mode**.
- The **interquartile range** is **$Q3 - Q1$** .
- The **median** is the **50th** percentile or **$Q2$** .
- **Variance** has units that are the **square** of the data units.
- **Standard deviation** has the **same** units as the data.
- A **perfect positive** correlation has **$r = 1$** ; perfect negative has **$r = -1$** .
- The **regression** line always passes through the point **$(\bar{x}, \bar{y})$** .
- The **product** of the two regression coefficients lies in **$[0, 1]$** and equals **r^2** .
- **Skewness** measures asymmetry; **kurtosis** measures **tailedness**.

29. Extended Rapid-Fire: Numerical Methods

- **Bisection** guarantees convergence but is **slow** (order 1).
- **Newton-Raphson** converges fastest (order 2) for a **simple** root.
- **Newton-Raphson** for $\sqrt{a}$ iterates **$x = (x + a/x)/2$** .
- The **fixed-point** iteration converges if **$|g'(x)| < 1$** .
- The **secant** method needs **2** initial points but **no** derivative.
- **Jacobi** and **Gauss-Seidel** are the **2** classic iterative linear solvers.
- **Gauss-Seidel** typically converges **faster** than **Jacobi**.
- **Diagonal dominance** is a **sufficient** condition for iterative convergence.
- **Partial pivoting** swaps rows to put the **largest** pivot on the diagonal.
- **Trapezoidal** rule approximates the area with **trapezoids** (straight tops).
- **Simpson's 1/3** fits **parabolas** through groups of **3** points.
- **Simpson's 3/8** fits cubics through groups of **4** points.
- **Romberg integration** refines the trapezoidal rule via **Richardson extrapolation**.
- **Euler's method** accumulates **global** error of order **$O(h)$** .
- **RK4** accumulates global error of order **$O(h^4)$** .
- The **step size h** controls the **accuracy** vs **cost** tradeoff.
- **Truncation error** comes from the **method**; **round-off error** from finite **precision**.

- **Newton's forward difference** interpolation suits **equally spaced** points near the start.
- **Lagrange interpolation** works for **unequally spaced** points.
- The **Newton-Raphson** convergence may **fail** near an **inflection** point or flat slope.

30. Extended Rapid-Fire: Cross-Topic Mix

- The **factorial** grows as **$O(n!)$** , faster than **exponential 2^n** .
- The **golden ratio $\phi \approx 1.618$** is the dominant Fibonacci root.
- **Euler's number $e \approx 2.71828$** is the base of natural logarithms.
- **$\pi \approx 3.14159$** relates a circle's circumference to its diameter.
- The **gcd** and **lcm** satisfy **$\text{gcd}(a,b) \cdot \text{lcm}(a,b) = a \cdot b$** .
- A number is divisible by **3** if its digit sum is divisible by **3**.
- The number of divisors of **$n = p^a q^b$** is **$(a+1)(b+1)$** .
- **Modular arithmetic** wraps values around a **modulus**.
- **Fermat's little theorem** states **$a^{p-1} \equiv 1 \pmod{p}$** for prime p .
- The sum **$1+2+\dots+n$** equals **$n(n+1)/2$** .
- The sum of the first n squares is **$n(n+1)(2n+1)/6$** .
- The sum of a geometric series is **$a(1-r^n)/(1-r)$** for **$r \neq 1$** .
- An infinite geometric series converges to **$a/(1-r)$** when **$|r| < 1$** .
- The **harmonic series $\sum 1/n$** **diverges**.
- **Logarithm** identity: **$\log(ab) = \log a + \log b$** .
- **Big-O** describes an **upper** bound; **Big-Ω** a **lower** bound; **Θ** a **tight** bound.

31. Extended Rapid-Fire: More Logic and Proof

- A **valid** argument has a true conclusion whenever all **premises** are true.
- A **sound** argument is valid **and** has all **true** premises.
- **Proof by contradiction** assumes the **negation** and derives a **false** statement.
- **Proof by contrapositive** proves **$\neg q \rightarrow \neg p$** instead of **$p \rightarrow q$** .
- **Mathematical induction** has **2** steps: the **base case** and the **inductive step**.
- **Strong induction** assumes the statement for **all** smaller cases.
- The **well-ordering principle** says every non-empty set of positive integers has a **least** element.
- A **counterexample** disproves a **universal** statement with a single instance.
- The **principle of explosion** says a **contradiction** implies **anything**.
- **Functional completeness**: **NAND** alone can express **all** Boolean functions.
- **NOR** alone is also **functionally complete**.
- There are **16** distinct binary Boolean functions of **2** variables.

32. Extended Rapid-Fire: More Sets and Relations

- The number of **antisymmetric** relations on n elements is **$2^n \cdot 3^{n(n-1)/2}$** .
- A relation can be neither **symmetric** nor **antisymmetric** simultaneously.
- The **composition** of two functions is associative: **$(f \circ g) \circ h = f \circ (g \circ h)$** .
- The **inverse** of a bijection f satisfies **$f \circ f^{-1} = \text{identity}$** .
- A **finite** set is always **countable**.
- The set of all **subsets** of $\mathbb{N}$ is **uncountable**.

- $|A \cup B \cup C|$ via inclusion-exclusion alternates $+, -, +$ signs.
- The **floor** function maps reals to the **largest integer** $\leq x$.
- The **ceiling** function maps reals to the **smallest integer** $\geq x$.
- A **partial function** may be **undefined** on some domain elements.

33. Extended Rapid-Fire: More Linear Algebra and Numerical

- A **homogeneous** system with more **unknowns than equations** has **infinite** solutions.
- The **dot product** of orthogonal vectors is **0**.
- The **cross product** magnitude equals the **parallelogram** area.
- The **norm** $\|v\| = \sqrt{v \cdot v}$.
- A set of vectors is **linearly independent** if the only zero combination is **trivial**.
- The **dimension** of a vector space is the size of its **basis**.
- The **column space** dimension equals the **rank**.
- **Eigenvectors** for distinct eigenvalues are **linearly independent**.
- The **spectral radius** is the **largest** absolute eigenvalue.
- A **stochastic** matrix has rows summing to **1** and largest eigenvalue **1**.
- The **power method** finds the **dominant** eigenvalue iteratively.
- **Cholesky** decomposition factors a **symmetric positive-definite** matrix as LL^T .
- **QR** decomposition factors $A = QR$ with **Q** orthogonal and **R** upper triangular.
- The **Frobenius norm** sums the **squares** of all entries.
- **Gauss-Jordan** elimination reduces to **reduced row-echelon** form.

34. Extended Rapid-Fire: More Calculus and Series

- The **Taylor series** expands f around a point using its **derivatives**.
- The **Maclaurin series** is a Taylor series centered at **0**.
- The Maclaurin series of e^x is $\sum x^n/n!$.
- The Maclaurin series of **sin x** has only **odd** powers.
- The Maclaurin series of **cos x** has only **even** powers.
- A **partial derivative** differentiates with respect to **one** variable, holding others fixed.
- The **gradient** points in the direction of **steepest ascent**.
- The **chain rule** for dy/dx via t is $(dy/dt)/(dx/dt)$.
- **L'Hopital** can be applied **repeatedly** while the form stays indeterminate.
- The **Wronskian** tests linear independence of **functions**.
- A **critical point** of a 2-variable function needs **both** partials zero.
- The **Hessian** determinant test classifies **2-variable** extrema.
- The **arc length** of a curve is $\int \sqrt{1 + (dy/dx)^2} dx$.
- The **volume of revolution** (disk method) is $\int \pi [f(x)]^2 dx$.

35. Extended Rapid-Fire: More Probability and Stats

- The **multiplication rule** for n independent events multiplies all $P(A_i)$.
- The **birthday paradox** gives **>50%** chance of a shared birthday with just **23** people.
- A **joint** distribution describes **2 or more** random variables together.
- The **marginal** distribution sums/integrates out the **other** variables.

- **Conditional expectation** $E[X|Y]$ is itself a **random variable**.
- The **moment generating function** encodes all **moments** of a distribution.
- The **first moment** about the origin is the **mean**.
- The **second central moment** is the **variance**.
- **Standardizing** a variable gives mean **0** and variance **1**.
- The **sample mean** is an **unbiased** estimator of the population mean.
- A **confidence interval** gives a range for a **parameter** at a stated confidence.
- **Type I error** rejects a true **null** hypothesis (false positive).
- **Type II error** fails to reject a false **null** hypothesis (false negative).
- The **p-value** is the probability of data at least as extreme under the **null**.
- The **normal** approximation to the binomial works when **np** and **n(1-p)** are ≥ 5 .

36. Extended Rapid-Fire: Final Mixed Set

- A **prime** number has exactly **2** divisors: **1** and itself.
- There are **25** primes below **100**.
- The number **2** is the only **even** prime.
- **Euclid's algorithm** computes the **gcd** by repeated **remainder**.
- The **base case** of recursion prevents **infinite** recursion.
- A **permutation matrix** has exactly one **1** per row and column.
- The **determinant** of a permutation matrix is ± 1 .
- The **logarithm base change** formula is $\log_b a = \ln a / \ln b$.
- **Stirling's approximation**: $n! \approx \sqrt{(2\pi n)}(n/e)^n$.
- The **number of edges** in a complete bipartite graph **K3,3** is **9**.
- The **chromatic number** of a tree with at least one edge is **2**.
- A **spanning tree** of a graph with n vertices always has **$n-1$** edges.
- The **trapezoidal rule** with **1** interval uses just the **2** endpoints.
- **Simpson's rule** needs at least **2** subintervals (3 points).
- The **order** of convergence of Newton-Raphson at a **double** root drops to **1** (linear).
- A **Markov chain** transition matrix has each row summing to **1**.
- The **steady-state** vector of a Markov chain is the eigenvector for eigenvalue **1**.

37. Bonus One-Liners

- The **rank-nullity theorem** states **rank + nullity = number of columns**.
- A **scalar matrix** is a diagonal matrix with all equal **diagonal** entries.
- The **minor** of an entry is the determinant of the submatrix after deleting its **row and column**.
- The **cofactor** is a minor with sign $(-1)^{(i+j)}$.
- A **Vandermonde** determinant is non-zero when all **nodes are distinct**.
- The **conditional probability chain rule** multiplies $P(A1)P(A2|A1)P(A3|A1A2)$.
- A **discrete uniform** over $1..n$ has mean $(n+1)/2$.
- The **Poisson** approximation improves as $n \rightarrow \infty$ with **np** fixed.
- The **trapezoidal error** is proportional to the **second derivative f''** .
- **Simpson's error** is proportional to the **fourth derivative $f^{(4)}$** .
- A **recurrence** of degree **k** needs **k** initial conditions.
- The **empty graph** on n vertices has **0** edges and **n** components.

- A **complete graph K_n** is **$(n-1)$** -regular.
- The **number of leaves** in a full binary tree with i internal nodes is **$i+1$** .
- The **determinant** of an orthogonal matrix is always **$+1$ or -1** .
- The **eigenvalues** of a projection matrix are **0 and 1** .